

Max–Min Fairness of CR-RSMA-Based UAV Relay-Assisted Emergency Communication Network With Limited User Energy

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Abstract—In post-disaster scenarios, it is challenging for affected users to transmit data as quickly as possible before their residual energy (RE) is exhausted. Besides, the problem of limited users' RE causes severe transmission delay unfairness within the network. In this article, we propose a novel two-phase scheme, called energy-aware unmanned aerial vehicle (UAV) relay transmission (EURT), to balance transmission delay of users and network fairness. Specifically, in the first phase, we pair users two-by-two based on their RE and minimize the maximum transmission delay among all pairs by jointly optimizing the bandwidth allocation, transmit powers, and UAV altitude. In the second phase, we design a cognitive radio (CR) inspired rate-splitting multiple access (RSMA) scheduling strategy to obtain the optimal power splitting factor for each pair. This strategy considers the user with a lower RE value in each pair as the primary user (PU) and the other one as the secondary user (SU), then minimizes the transmission delay of the SU while ensuring the Quality of Service (QoS) of the PU. Furthermore, we propose a novel evaluation framework to explore the degree of impact of RE and channel state information (CSI) on network delay fairness. Simulation results demonstrate that: 1) the proposed EURT algorithm effectively improves performance metrics of networks in terms of transmission delays, throughput, energy consumption and energy efficiency and 2) the proposed algorithm achieves a tradeoff between the minimum delay and the optimal network fairness by adjusting the QoS threshold of the PU.

Index Terms—CR-RSMA, emergency communication, resource optimization, unmanned aerial vehicle (UAV) relay, users energy limited.

Manuscript received 14 December 2023; revised 19 February 2024 and 4 April 2024; accepted 8 April 2024. Date of publication 17 April 2024; date of current version 26 June 2024. This work was supported in part by the Joint Fund for Regional Innovation and Development of the National Natural Science Foundation of China under Grant U21A20445; in part by the National Natural Science Foundation of China under Grant 62201224; in part by the Jilin Province Development and Reform Commission Project under Grant 2023C039-1; and in part by the Jilin Provincial Key Laboratory of Intelligent Sensing and Network Technology under Grant 20240302096GX, Grant 20230508035RC, and Grant YDZJ202102CXJD018. (Corresponding authors: Fengye Hu; Zhuang Ling.)

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Digital Object Identifier 10.1109/JIOT.2024.3390212

I. INTRODUCTION

IN THE face of disasters, such as tsunamis, earthquakes, floods, hurricanes, etc., communication infrastructure is often completely or partially destroyed, leaving affected areas without access to the outside world [1], [2]. It is a crucial task to temporarily restore connections and provide high-capacity, low-latency, and high-reliability communication services to users' equipment or Internet of Things (IoT) devices after a disaster [3], [4]. To accommodate the characteristics of low cost, easy deployment, and rapid response of emergency communication networks, it is a promising way to use the unmanned aerial vehicle (UAV) to establish a wireless relay network to achieve information interaction between affected areas and the external environment [5]. Specifically, UAV relay-assisted emergency communication services have the following advantages: 1) the UAV has a greater chance of generating Line-of-Sight (LoS) transmission conditions, which can help improve the user's Quality of Service (QoS) by reducing signal attenuation [6]; 2) since UAVs have the advantages of flexible deployment, fast response, high mobility, etc., it is more suitable for wireless communication systems with on-demand dynamic traffic tasks [7]; and 3) compared to fixed infrastructures, UAVs can dynamically adjust their spatial position to be closer to ground users, thus improving the transmission quality of wireless channels [8].

With the ever-changing communication capability requirements of emergency communications operations, there are still some challenges in UAV-assisted emergency communications networks. Since critical ground infrastructure, such as charging facilities, is vulnerable to damage from secondary disasters, the residual energy (RE) of affected users (AUs) and UAVs is usually limited [9], [10], [11]. In addition, considering the complex geographical conditions of the disaster area and the scattered distribution of users, there is an obvious *near-far* effect among different users [12]. When the RE of edge users is limited, the decrease of radio frequency energy will inevitably aggravate the *near-far* effect, leading to serious fairness issues. Therefore, users' data should be transmitted as quickly as possible before its RE is exhausted while ensuring the fairness of the network.

However, there is a contradiction between minimizing users' transmission delay and maximizing network fairness.

Since the available communication resources provided to the emergency communication network are usually extremely limited, improving users' performance in the network will inevitably require more system resources to be tilted toward them, resulting in unfair resource allocation. Therefore, how to design a transmission scheme to achieve the tradeoff between user performance and network fairness in UAV relay-assisted emergency communications is very important.

A. Motivations and Related Works

Utilizing the flexibility of UAVs and carrying communication modules as aerial relay nodes could realize rapid reconfiguration of the network [13], [14], [15], [16]. Aiming to maximize the throughput of the UAV relay system, Zeng et al. [13] proposed an iterative algorithm to alternately optimize power allocation and UAV relay trajectories. To enhance the relay system reliability, Chen et al. [14] investigated the optimal height and location of UAV relays using a realistic channel model. Reliability indicators, such as power loss, outage probability, and bit error rate, were focused on. Hu et al. [15] investigated the fairness problem of UAV-assisted relay networks in urban areas. 3-D UAV trajectories, users' scheduling, and bandwidth allocation are jointly optimized to maximize the minimum throughput for all users. Liu et al. [16] investigated a relay system using multiple UAV links to communicate simultaneously with the destination via the spectrum-sharing of multiple UAV relays. They jointly optimized 3-D UAV trajectories, and transmit powers of source nodes and UAVs to maximize the sum throughput of all links.

Recently, more and more work has focused on deploying UAVs as aerial relays to provide emergency communication services in remote areas or disaster areas due to the obvious LoS probability advantage [17], [18], [19]. Hu et al. [17] investigated an emergency communication network in an urban environment, in which a drone provided wireless relay connections to multiple disconnected ground wireless access points in a half-duplex mode. Considering the rate fairness constraints among multiple nodes, the hovering altitude, transmit power and bandwidth allocation of the drone are jointly optimized to maximize the total throughput of the system. Yang et al. [18] studied the problem of multi-UAV-assisted IoT network data transmission in disaster areas and proposed a time-aware trajectory optimization strategy. While considering the energy budget of the drone and the data requirements of the IoT nodes, a balance between the energy consumption (EC) of the drone and the task completion time is achieved. Wang et al. [19] investigated a UAV relay-assisted emergency communication network with half-duplex downlink transmission. Considering that global channel state information (CSI) is difficult to obtain by decision makers in advance, the author proposes a strategy based on Q-learning and convolutional neural networks to solve a joint macro base station power allocation and optimal UAV service area selection problem. However, most existing work on emergency communications only considers the situation where UAV battery power is limited, ignoring the impact on network transmission delay when both users and UAV energy are limited in disaster scenarios.

In order to further enhance the spectrum resource utilization, an increasing number of research are focusing on applying nonorthogonal multiple access (NOMA) transmission protocols in UAV communication systems [20], [21], [22], [23]. Zhai et al. [20] investigated the combination of NOMA technology and UAV relay in an emergency communication network. To fully explore the advantages of NOMA-DF-relay, they jointly optimized the UAV's altitude, users' channel allocation and UAV's transmit power. Do et al. [21] studied a UAV relay network with radio frequency energy harvesting, in which the drone harvested energy from the signal transmitted by the BS and used NOMA to forward the message to users. To evaluate network performance, they derived closed-form expressions of both outage probability and ergodic capacity under Rician fading channel conditions. Tang et al. [22] investigated a physical layer security strategy for UAV communication systems combining NOMA and cognitive radio (CR), i.e., CR-NOMA. 3-D drone trajectories and power allocation are jointly optimized to maximize the worst-case average secrecy sum rate for all secondary users (SUs) without negatively affecting the performance of the primary users (PUs). All the aforementioned works do not fully explore the network fairness issues caused by user energy constraints in emergency communications while focus merely on maximizing the network throughput. First, NOMA usually prioritizes decoding users with better CSI to ensure network fairness. However, when users are close to each other and their RE gaps are high, the traditional CSI-based scheduling do not guarantee fairness and increase user EC. Second, since the pattern of the NOMA protocol causes the earliest decoded user to suffer the most interference from all the remaining users, this user requires a larger transmit power to maintain network fairness. When users' RE is limited, the fairness of the network may be further degraded.

Recently, rate-splitting multiple access (RSMA) has attracted considerable attention since it can achieve any point of the multiple access channel (MAC) capacity region boundary, i.e., it has better fairness than NOMA [24], [25], [26]. Yang et al. [27] investigated an uplink RSMA wireless communication system intending to maximize users' sum rate ensuring a proportional fairness constraint by jointly optimizing decoding orders at the receiver and users' transmit powers. Although using RSMA can achieve fairer multiuser access capabilities than NOMA [28], spectrum resources and energy in emergency communication networks are usually extremely limited and it is an unnecessary waste for all users to always occupy the same radio resource block (RRB). Besides, due to the lack of energy supplement avenues, users with lower RE must transmit their data as quickly as possible, while users with higher RE have relatively lower transmission priorities. Fortunately, the goal of CR is to utilize the spectrum efficiently. When the SUs do not negatively affect the PUs, CR opportunistically enables low-priority users to share the resource blocks originally allocated to the high-priority users [29], [30]. Therefore, it is expected to further improve the spectrum utilization of the network by combining CR and RSMA, with currently limited research in this area. Xiao et al. [31] studied an uplink network based on two

CR-inspired RSMA and SIC protocols. The authors derived the SU's ergodic rate considering the PU's specific QoS for both protocols and compared the two protocols qualitatively and quantitatively.

Current, research on integrating CR-RSMA with UAVs have not been extensively explored, and some key issues remain. First, most existing studies assume predefined resource allocation methods without considering optimization design, which makes it difficult to apply in UAV communication networks due to severe Air-to-Ground (AtG) link interference and spectrum scarcity. Second, the aforementioned works have not considered how to balance delay and fairness in energy-limited emergency communication networks. Motivated by this, we attempt to eliminate the network fairness degradation problem due to limited users' RE and provide a potential possibility to tradeoff optimality and fairness. As distinguished from existing research, our work considers the RE constraints of both users and the UAV as well as the impact of limited user energy on network fairness.

B. Contributions and Organization

In this article, we investigate the max-min fairness problem caused by limited users' RE in emergency communication networks. A novel energy-aware UAV relay transmission strategy, called EURT, is proposed to balance minimum transmission delay and optimal network fairness. This strategy includes a two-stage optimization process of resource allocation and user access scheduling. The major contributions are summarized as follows:

- 1) We propose a new emergency communication network framework based on UAV relay. When communication and energy supply infrastructures are damaged, UAVs can forward users' information to nearby available BS promptly. We formulate a max-min fairness optimization problem by jointly optimizing the bandwidth allocation of each RRB, the UAV's altitude, the transmit power of the UAV and users, as well as the power splitting coefficient of each pair. The QoS threshold constraint of the PU causes the problem to be nonconvex.
- 2) We first transform the original problem into a maximizing minimum two-user capacity region problem, thereby eliminating the power splitting coefficient as an optimization variable. Second, we adopt the block coordinate descent (BCD) algorithm to decompose the problem into resource allocation and altitude optimization subproblems. Finally, we iteratively solve the two subproblems using the successive convex approximation (SCA) algorithm until convergence.
- 3) After obtaining the optimal transmission power for each user, we formulate a user access scheduling strategy based on the CR-RSMA transmission paradigm. This strategy maximizes the transmission delay fairness for each pair of users while ensuring the QoS constraints of the PU and derives the closed-form expression of the power splitting coefficient for each pair of users.

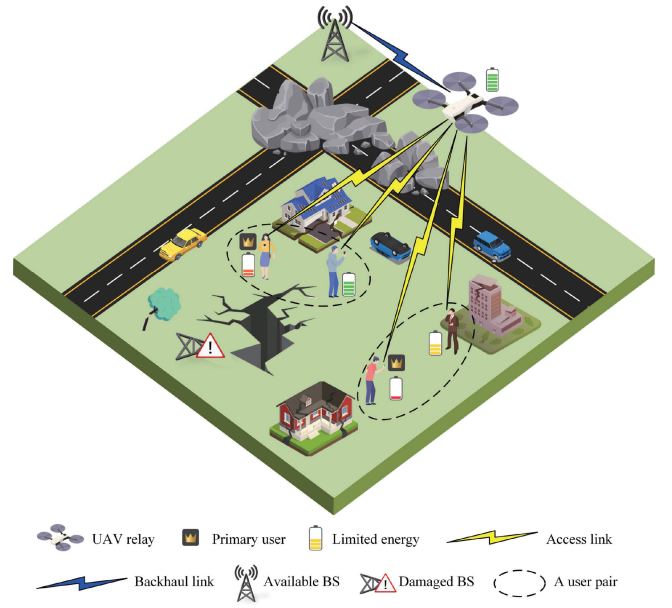


Fig. 1. UAV relay-assisted emergency communication network.

- 4) We verified the effectiveness of the proposed EURT algorithm. The simulation results show that compared with other benchmark algorithms, the EURT algorithm achieves better performance in terms of delay, throughput, EC and EE, and can achieve a tradeoff between user transmission delay and network fairness.
- 5) We design a novel evaluation framework to investigate the impact of two scheduling strategies on network fairness. The results indicate that when the gap in RE between users exceeds a threshold, the RE-aware strategy has a greater impact on fairness than the CSI-aware strategy. Conversely, the CSI-aware strategy performs better in terms of fairness when the RE gap is below the threshold.

The remainder of this article is organized as follows. The system model and optimization problem are formulated in Section II. In Section III, we describe the proposed EURT strategy. Simulation results are provided in Section IV. Finally, Section V concludes this article and suggests some possible future research work.

II. SYSTEM MODEL AND PROBLEM FORMULATION

As shown in Fig. 1, an emergency communication network consisting of M AUs, a rotary-wing UAV, and a nearby available BS is considered, where the BS initially deployed in the disaster area to support wireless backhaul is damaged and cannot provide wireless information backhaul for users within its coverage area. Besides, users' devices have limited energy due to disaster damage to the charging unit, and need to transmit data back to the command and control center before their power is exhausted. To achieve fast network recovery and take advantage of the AtG channel propagation, a rotary-wing UAV is deployed to relay crucial messages from AUs to the nearby available BS, where direct links from AUs to the BS are considered unavailable. The UAV adopts a

TABLE I
SUMMARY OF THE MAIN NOTATIONS

Notation	Definition
B	Total available bandwidth
h_U	UAV altitude
x_U, y_U	The horizontal coordinate of UAV
x_B, y_B	The horizontal coordinate of BS
d_{iU}	Horizontal distance between user i and UAV
d_{UB}	Horizontal distance between UAV and BS
a_0, b_0	Environmental parameters
θ_{ij}	Elevation angle in degrees for link i - j
$P_{LoS,ij}$	LoS probability for link i - j
$P_{NLoS,ij}$	NLoS probability for link i - j
η_{LoS}	Additional path loss for LoS link i - j
η_{NLoS}	Additional path loss for NLoS link i - j
β_0	Reference channel gain per unit distance
D_{ij}	Euclidean distance between nodes i and j
M	The number of users
A_i	The i -th user
Γ_{iU}	The Channel gain between user i and UAV
Γ_{UB}	The Channel gain between UAV and BS
N	Number of user pairs
$m_{n,j}$	Packet length for the j -th user in the n -th pair
$E_{n,j}$	Remaining energy for the j -th user in the n -th pair
$P_{n,j}$	Transmit power of the j -th user in the n -th pair
P_U	Transmit power of UAV
T_{th}	Maximum transmission delay for each user pair
N_0	Noise power spectral density
$R_{n,U}$	Sum achievable rate between n -th pair of users to UAV
$R_{n,B}$	Sum achievable rate between UAV to BS
ω_n	Bandwidth allocation factor for the n -th user pair
α_n	Power allocation factor for the n -th pair of users
$R_{n,th}$	QoS requirements for the primary user in the n -th pair

decode-and-forward (DF) protocol and equips dual antennas, one for receiving information from the uplink of users in the disaster area, and the other for forwarding the backhaul link information to the available BS. To further enhance the utilization of wireless resources, the full-duplex (FD) relay mode is adopted. In this approach, two-hop links in the relay network occupy the same subcarrier, thus introducing self-interference at the relay. For convenience analysis, we summarize significant notations in Table I.

A. Air-to-Ground Channel Model

Assume that M energy-constrained users are in the affected area, and that M is an even number. The location of i th AU is denoted as $L_i = (x_i, y_i, 0) \quad \forall i \in \mathcal{M} \triangleq \{1, 2, \dots, M\}$. Let (x_U, y_U, h_U) be the coordinate of the UAV. The BS is located at the coordinate $(x_B, y_B, 0)$. As a result, the horizon distance between AU_i and the UAV is determined by

$$d_{iU} = \sqrt{(x_U - x_i)^2 + (y_U - y_i)^2} \quad \forall i \in \mathcal{M} \quad (1)$$

and the horizon distance between the UAV and the BS is

$$d_{UB} = \sqrt{(x_U - x_B)^2 + (y_U - y_B)^2}. \quad (2)$$

According to the statistical channel model of path loss of low-altitude platforms in different urban environments proposed in [32], it can be seen that the AtG channel is mainly composed of free space path loss and additional path loss with different propagation groups (LoS or NLoS). Furthermore, the

LoS occurrence probability between the transmitter i and the receiver j is a function of the elevation angles as the following:

$$P_{ij}^{LoS} = a_0(\theta_{ij} - \theta_0)^{b_0} \quad (3)$$

where $\theta_0 = 15^\circ$ is the minimum angle allowed by the system model. a_0 and b_0 are environmental parameters. θ_{ij} is the elevation angle in radians, where $\theta_{ij} = (180/\pi)\tan^{-1}(h_U/d_{ij})$. To ensure a nonnegative LoS probability, θ_{ij} is higher than θ_0 . On the other hand, the NLoS occurrence probability between the transmitter i and the receiver j can be simply calculated as $P_{ij}^{NLoS} = 1 - P_{ij}^{LoS}$. In addition, Hourani et al. [32] considered that the statistical characteristics of the additional path loss obey a Gaussian distribution, whose mean is a constant that depends on the environment, and the standard deviation is a function of the AtG link elevation angle. Define $\mathcal{N}(\mu_{LoS}, \sigma_{LoS}^2(\theta_{ij}))$ and $\mathcal{N}(\mu_{NLoS}, \sigma_{NLoS}^2(\theta_{ij}))$ represent the statistical Gaussian distribution characteristics in the LoS transmission state and the NLoS transmission state, respectively. The average channel gain of the AtG link based on the probability of LoS transmission can be modeled as

$$\Gamma_{ij} = h_{ij}g_{ij} \quad (4)$$

where $g_{ij} = \beta_0 D_{ij}^{-2}$ represents the free space path loss, β_0 is the reference channel gain per unit distance, and $D_{ij} = \sqrt{d_{ij}^2 + h_U^2}$ represents the Euclidean distance between nodes i and j . For simplicity, $h_{ij} = \eta_{LoS}P_{ij}^{LoS} + \eta_{NLoS}P_{ij}^{NLoS}$ represents the additional path loss of LoS transmission probability and, where $\eta_{LoS} = 10^{-0.1\mu_{LoS}}$ and $\eta_{NLoS} = 10^{-0.1\mu_{NLoS}}$ are taken as the mean of the corresponding Gaussian distribution, respectively. Therefore, Γ_{ij} can be further written as

$$\begin{aligned} \Gamma_{ij} &= (\eta_{LoS}P_{ij}^{LoS} + \eta_{NLoS}P_{ij}^{NLoS})\beta_0 D_{ij}^{-2} \\ &= (\sigma_{NLoS} + \sigma_{LoS}P_{ij}^{LoS})D_{ij}^{-2} \end{aligned} \quad (5)$$

where expressions $\sigma_{NLoS} = \eta_{NLoS}\beta_0$ and $\sigma_{LoS} = (\eta_{LoS} - \eta_{NLoS})\beta_0$ represent additional path loss for NLoS and LoS propagation, respectively.

B. Energy-Prioritized CR-RSMA Scheduling Scheme

We consider terrestrial users using the underlying CR paradigm (i.e., concurrent transmission of PU and SU is allowed as long as the interference of SU to PU does not exceed a tolerable threshold) to access available RRBs [33]. The BS establishes a standard to determine whether a user is eligible for transmission and broadcasts the access scheduling scheme to all users through UAV relays, thereby controlling multiple access. Since the AtG channel is a statistical model based on LoS propagation probability, which is only related to the propagation parameters, elevation angle and distance. Therefore, the BS can obtain the average CSI as follows: 1) before the UAV starts relay transmission, AUs forward their RE state and location information to BS via the control and nonpayload communication (CNPC) links relayed by the UAV and 2) the BS estimates the propagation coefficients using the collected user data to accurately obtain channel gains of AtG links from AUs to the UAV; and 3) the BS formulates an

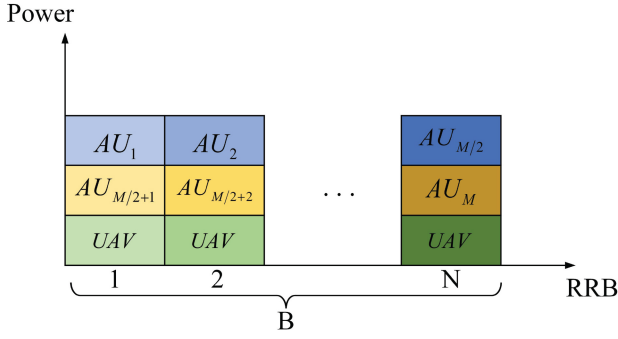


Fig. 2. Multi-RRB allocation in the relay system.

appropriate user access scheduling scheme and forwards it to AUs through the UAV using CNPC links.

This scheduling scheme groups all the AUs into $N = M/2$ pairs according to the value of their RE, and each pair consists of two users as $\{A_1, A_{M/2+1}\}, \{A_2, A_{M/2+2}\}, \dots, \{A_{M/2}, A_M\}$, where their RE are in sequence $E_1 \leq E_2 \leq \dots \leq E_{M/2} \leq \dots \leq E_M$. For each pair, we set the user with the lower RE as the PU and the other as the SU. The total available bandwidth B is divided into N orthogonal RRBs, where different pairs are allocated different RRBs for transmission. Let ω_n denote the bandwidth allocation factor of the n th RRB, i.e., users of the n th pair share the RRB with bandwidth $\omega_n B$ for information transmission, as illustrated in Fig. 2. In particular, this scheduling scheme ensures that the first $M/2$ users with smaller REs are always PUs and prioritizes their performance. Besides, each pair of users can be decoded separately in different frequency bands using parallel SIC decoding, thus reducing the decoding overhead.

According to a two-user uplink RSMA approach, only one user needs to split its messages to achieve the full pentagon of the capacity region. To minimize the interference to the PU, we split the SU's message into two data streams. The message received by the UAV relay from the n th pair of users can be expressed as

$$y_{n,U} = \sqrt{\alpha_n P_{n,S}} \Gamma_{n,SU} x_{S1} + \sqrt{(1 - \alpha_n) P_{n,S}} \Gamma_{n,SU} x_{S2} + \sqrt{P_{n,P}} \Gamma_{n,PU} x_P + n_U \quad (6)$$

where $P_{n,P}$ and $P_{n,S}$ represent transmit powers of PU and SU, respectively. α_n denotes the power splitting factor of the n th pair. n_U is the additive white Gaussian noise at the UAV relay with zero mean and variance σ^2 . The message x_{S1} represents the first data stream of the SU, which is decoded at the receiver by treating x_P and x_{S2} as interference and then removing them from the received signal. The message x_P is decoded second at the receiver and treats x_{S2} as interference. The message x_{S2} is finally decoded without treating any other user's signal as interference. Therefore, the achievable rates for the primary and SUs of the n th pair can be expressed as follows:

$$R_{n,S} = \omega_n B \log_2 \left(1 + \frac{\alpha_n \gamma_{n,S}}{\gamma_{n,P} + (1 - \alpha_n) \alpha_n \gamma_{n,S} + 1} \right) + \omega_n B \log_2 \left(1 + (1 - \alpha_n) \alpha_n \gamma_{n,S} \right) \quad \forall n \quad (7)$$

and

$$R_{n,P} = \omega_n B \log_2 \left(1 + \frac{\gamma_{n,P}}{(1 - \alpha_n) \alpha_n \gamma_{n,S} + 1} \right) \quad \forall n. \quad (8)$$

For two users in the n th pair, define $\gamma_{n,i} \triangleq (P_{n,i} \Gamma_{n,iU} / \omega_n N_0 B)$, $i \in \{P, S\}$ as the normalized signal-to-noise ratio of PU and SU, and the QoS threshold of PU is $R_{n,th}$. Thus, we obtain the constraints on the QoS of PU as follows:

$$\mathcal{C}_1 : R_{n,P} \geq R_{n,th} \quad \forall n. \quad (9)$$

The transmit delay for the SU in n th user pair is given as

$$t_{n,S} = \frac{m_{n,S}}{R_{n,S}} \quad \forall n \quad (10)$$

where $m_{n,S}$ is the packet length to be transmitted by the SU of the n th pair.

The link that forwards the traffic of the n th user pair through the UAV to the available BS is denoted as $n - u - b$. Define the UAV's transmit power as P_U and the RE of the j th user in the n th pair is $E_{n,j}$. Since UAV uses the FD relay mode and access and backhaul links share the same RRB, the achievable rates of the two-hop links are [34]

$$R_{n,U} = R_{n,P} + R_{n,S} \quad \forall n \quad (11)$$

and

$$R_{n,UB} = B \omega_n \log_2 \left(1 + \frac{P_U \Gamma_{UB}}{B \omega_n N_0} \right) \quad \forall n \quad (12)$$

where ω_n represents the bandwidth allocation factor of link $n - u - b$, N_0 is the noise power spectral density, and $B \omega_n N_0$ is the additive Gaussian noise; SI represents the self-interference cancellation coefficient that reflects the self-interference cancellation capability, and P_U/SI represents the residual self-interference from the second hop of link $n - u - b$.

Therefore, we can obtain the following expression for the achievable rate of the link $n - b$ based on the capacity theory of the relay system [35]:

$$R_{n,B} = \min\{R_{n,U}, R_{n,UB}\}. \quad (13)$$

Furthermore, The user's EC mainly depends on the transmission energy [36], while the UAV's EC mainly consists of transmission and hovering energy. We define the maximum transmission delay of the system as T_{th} . Therefore, EC constraints of users and UAV can be obtained, respectively, as follows:

$$\mathcal{C}_2 : P_{n,j} T_{th} \leq E_{n,j}^{th} \quad \forall n \in \left\{ 1, \dots, \frac{M}{2} \right\}, j \in \{P, S\} \quad (14)$$

and

$$\mathcal{C}_3 : (P_U + P_0 + P_i) T_{th} \leq E_U^{th} \quad (15)$$

where P_0 and P_i are two constants representing the blade profile and the induced power when hovering, respectively [37].

C. Problem Formulation

Aiming at minimizing the transmission delay for low-RE users and ensuring the fairness of the network, the max-min fairness metric is introduced, which enables a higher level of fairness by improving the performance of the worst-case users [38]. To achieve max-min fairness without negatively affecting PU performance, we minimize the maximum transmission delay of the SU while ensuring QoS for the PUs among all pairs by jointly optimizing bandwidth allocation coefficients, UAV's altitude, users and UAV's transmit power, and SU's power splitting factor in each pair. The optimization problem of minimizing the maximum transmission delay is formulated as

$$\begin{aligned}
 (\mathbf{P1}) : \quad & \min_{\omega_n, h_U, P_{n,j}, P_U, \alpha_n} \max_n t_{n,S} \\
 \text{s.t.} \quad & C_1, C_2, C_3 \\
 & C_4 : 0 \leq \alpha_n \leq 1 \quad \forall n \\
 & C_5 : \sum_{n=1}^N \omega_n \leq 1 \\
 & C_6 : 0 \leq \omega_n \leq 1 \quad \forall n \\
 & C_7 : h_U \geq h_{\min}
 \end{aligned} \tag{16}$$

where the constraint C_1 indicates that the achievable rate of PU cannot be lower than its QoS threshold in the n th pair, which guarantees the performance of data transmission. Constraint C_2 and C_3 indicate that users and UAV's EC cannot exceed their maximum energy limits, respectively. Constraint C_4 denotes the power splitting constraint for the SU, i.e., the total transmit power of two split information streams is not greater than that of the original user in the n th pair. C_5 and C_6 represent bandwidth allocation constraints, i.e., users in each pair share an orthogonal RRB, and the total bandwidth of all RRBs is B . Constraint C_7 indicates that the UAV cannot fly below a minimum altitude to ensure LoS channel conditions.

Remark 1: By using RSMA, all points on the boundary of the two-user ($u1$ and $u2$) capacity region can be achieved. We assume that the message of $u2$ x_{u2} is split into two information streams x_{u21} and x_{u22} , respectively, and the data of $u1$ is not split. All possible SIC decoding orders can be expressed as $x_{u21} \rightarrow x_{u22} \rightarrow x_{u1}$, $x_{u1} \rightarrow x_{u21} \rightarrow x_{u22}$, and $x_{u21} \rightarrow x_{u1} \rightarrow x_{u22}$. Defining $\eta_i = (P_i \Gamma_i / \sigma^2)$, $i \in \{x_{u1}, x_{u21}, x_{u22}\}$ and the achievable rate pair obtained by the decoding order $x_{u21} \rightarrow x_{u22} \rightarrow x_{u1}$ is $R_{u21} = \log_2(1 + [\eta_{u21}/(\eta_{u22} + \eta_{u1} + 1)])$, $R_{u22} = \log_2(1 + [\eta_{u22}/(\eta_{u1} + 1)])$, and $R_{u1} = \log_2(1 + \eta_{u1})$. The sum achievable rate of $u2$ is $R_{u2} = R_{u21} + R_{u22} = \log_2(1 + [(\eta_{u21} + \eta_{u22})/(\eta_{u1} + 1)])$, and the achievable rate of $u2$ obtained by NOMA using decoding order $x_{u2} \rightarrow x_{u1}$ is $R_{u2}^{\text{NOMA}} = \log_2(1 + [\eta_{u2}/(\eta_{u1} + 1)])$. Since RSMA can achieve any point on the boundary of the capacity domain by adjusting the power splitting factor, and the sum achievable rate of these points is the same [28], decoding order $x_{u21} \rightarrow x_{u22} \rightarrow x_{u1}$ ($x_{u1} \rightarrow x_{u21} \rightarrow x_{u22}$) is equivalent to decoding order $x_{u2} \rightarrow x_{u1}$ ($x_{u1} \rightarrow x_{u2}$). In addition, the achievable rate pairs $\{R_{u2}, R_{u1}\}$ and $\{R_{u1}, R_{u2}\}$ obtained through the two decoding sequences of NOMA cannot

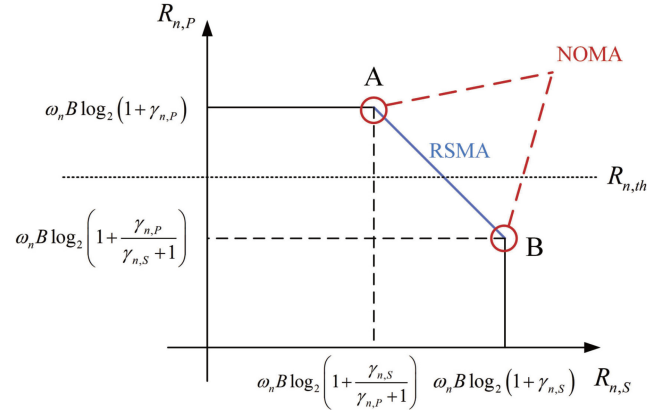


Fig. 3. Two-user uplink RSMA communication capacity region.

achieve the complete uplink MACs capacity region boundary. Therefore, for the RSMA transmission scheme, decoding order $x_{u21} \rightarrow x_{u1} \rightarrow x_{u22}$ can achieve the performance of the entire capacity region boundary.

III. PROPOSED OPTIMIZATION

In this section, we propose an efficient two-stage algorithm to solve the nonconvex problem (P1). Specifically, inspired by the fact that RSMA can achieve the same optimal capacity region corner point as NOMA [31], the achievable capacity domain of two-user system of the RSMA scheme is

$$C_{\text{RSMA}} = \left\{ (R_1, R_2) : R_1 + R_2 \leq \log_2 \left(1 + \frac{P_1 + P_2}{N_0} \right) \right\} \tag{17}$$

where

$$R_1 \leq \log_2 \left(1 + \frac{P_1}{N_0} \right), R_2 \leq \log_2 \left(1 + \frac{P_2}{N_0} \right). \tag{18}$$

Equation (17) portrays the area enclosed by a pentagon, which corresponds to the capacity domain of the n th pair of users adopting RSMA transmissions of our proposed problem. As shown in Fig. 3, NOMA can only achieve two corner points of the capacity region boundary, while RSMA can achieve all boundary points. For each user pair, $R_{n,P}$ and $R_{n,S}$ may achieve optimal fairness only if their achievable capacity region lie on the connection between points A and B. Therefore, the original problem is first transformed into the problem of maximizing the minimum two-user capacity region under RSMA transmission and then find the power allocation coefficients that ensure QoS constraints and optimal fairness. On the other hand, in order to satisfy the constraint C_1 , the line represented by QoS threshold $R_{n,th}$ of the PU should pass through the pentagon domain of the capacity region. When $0 \leq R_{n,th} \leq \omega_n B \log_2(1 + [\gamma_{n,P}/\gamma_{n,S} + 1])$, the maximum SU achievable rate is obtained when the SU is not split and decoded last to satisfy the constraint C_1 . This case has the worst fairness, so we consider the following constraint:

$$C_{1'} : \omega_n B \log_2 \left(1 + \frac{\gamma_{n,P}}{\gamma_{n,S} + 1} \right) \leq R_{n,th} \quad \forall n \tag{19}$$

and

$$C_{1''} : \omega_n B \log_2(1 + \gamma_{n,P}) \geq R_{n,th} \quad \forall n. \tag{20}$$

Then, the problem (P1) can be reformulated as follows:

$$\begin{aligned}
 (\mathbf{P2}) : \quad & \max_{\omega_n, h_U, P_{n,j}, P_U} \min_n R_{n,B} \\
 \text{s.t.} \quad & C_{1'}, C_{1''}, C_2, C_3 \\
 & C_5 \sim C_7.
 \end{aligned} \quad (21)$$

Since the objective function of this optimization problem (P2) is in discrete form, and it is not a convex optimization problem, we transform the objective function into continuous form by introducing a slack variable ξ and adding a constraint that ensures the achievable rate of link $n-b$ is not less than ξ . The reformulated optimization problem is shown as follows:

$$\begin{aligned}
 (\mathbf{P3}) : \quad & \max_{\omega_n, h_U, P_{n,j}, P_U, \xi} \xi \\
 \text{s.t.} \quad & C_0: \xi \leq R_{n,B} \quad \forall n \\
 & C_{1'}, C_{1''}, C_2, C_3, \\
 & C_5 \sim C_7.
 \end{aligned} \quad (22)$$

In addition, (P3) is still a nonconvex problem since constraints C_0 , $C_{1'}$ and $C_{1''}$ are nonconvex. For the constraint C_0 , it can be rewritten as

$$\begin{aligned}
 C_{01} : \xi &\leq R_{n,U} \quad \forall n \\
 C_{02} : \xi &\leq R_{n,UB} \quad \forall n.
 \end{aligned} \quad (23)$$

We divide the problem into two subproblems using the BCD algorithm and solve these two subproblems using the SCA technique, respectively.

A. Resource Allocation Subproblem

We fix the UAV flight altitude h_U , and problem (P3) can be transformed into a more simplified optimization problem

$$\begin{aligned}
 (\mathbf{P4}) : \quad & \max_{\omega_n, P_{n,j}, P_U, \xi} \xi \\
 \text{s.t.} \quad & C_{01}, C_{02}, C_{1'}, C_{1''}, C_2, C_3, C_5, C_6 \quad (24)
 \end{aligned}$$

where constraints C_{01} , C_{02} , $C_{1'}$ are nonconvex with respect to the variable ω_n , $P_{n,j}$. We use SCA methods to deal with these nonconvex constraints. First, we introduce auxiliary variables $\eta_{n,U}$ and $\eta_{n,B}$ to make constraints C_{01} and C_{02} satisfy

$$\xi \leq \omega_n B \eta_{n,U} \quad \forall n \quad (25)$$

$$\eta_{n,B} \leq \log_2 \left(1 + \frac{\sum_{j \in \{P,S\}} P_{n,j} \Gamma_{n,jU}}{\frac{P_U}{SI} + B\omega_n N_0} \right) \quad \forall n \quad (26)$$

and

$$\xi \leq \omega_n B \eta_{n,B} \quad \forall n \quad (27)$$

$$\eta_{n,B} \leq \log_2 \left(1 + \frac{P_U \Gamma_{UB}}{B\omega_n N_0} \right). \quad (28)$$

Constraint (27) can be equivalently transformed into

$$\ln(\xi) \leq \ln(\omega_n B) + \ln(\eta_{n,U}). \quad (29)$$

The first-order Taylor expansion (FTE) of $\ln(\xi)$ at the r th iteration is its convex upper bound

$$\ln(\xi) \leq \ln(\xi^r) + \frac{(\xi - \xi^r)}{\xi^r} \triangleq f_{\xi}^r. \quad (30)$$

The expression on the right-hand side (RHS) of (26) is convex with respect to P_U , ω_n , and its concave lower bound can be expressed as

$$\begin{aligned}
 & \log_2 \left(1 + \frac{\sum_{j \in \{P,S\}} P_{n,j} \Gamma_{n,jU}}{\frac{P_U}{SI} + B\omega_n N_0} \right) \\
 & \geq \log_2 \left(\frac{P_U}{SI} + B\omega_n N_0 + \sum_{j \in \{P,S\}} P_{n,j} \Gamma_{n,jU} \right) \\
 & \quad - \log_2 \left(\frac{P_U^r}{SI} + B\omega_n^r N_0 \right) \\
 & \quad - \frac{1}{\ln 2} \frac{N_0 B (\omega_n - \omega_n^r) SI + P_U - P_U^r}{P_U^r + B\omega_n^r N_0 SI} \triangleq f_{\omega_n, U}^r. \quad (31)
 \end{aligned}$$

Similarly, (27) and (28) can be reformulated by using FTE of ω_n at the r th iteration as

$$\ln(\xi) \leq \ln(\omega_n B) + \ln(\eta_{n,B}) \quad (32)$$

and

$$\begin{aligned}
 & \log_2 \left(1 + \frac{P_U \Gamma_{UB}}{B\omega_n N_0} \right) \geq \log_2 (B\omega_n^r N_0 + P_U \Gamma_{UB}) \\
 & \quad - \log_2 (B\omega_n^r N_0) - \frac{(\omega_n - \omega_n^r)}{\omega_n^r \ln 2} \triangleq f_{\omega_n, B}^r. \quad (33)
 \end{aligned}$$

The expression on the left-hand side (LHS) of constraint $C_{1'}$ is nonconvex with respect to the variable $P_{n,P}$, and we transform constraint $C_{1'}$ by introducing an auxiliary variable φ into the following form:

$$\ln(\omega_n B) + \ln(\varphi) \leq \ln(R_{n,th}) \quad (34)$$

and

$$\begin{aligned}
 \varphi & \geq \log_2 \left(1 + \frac{P_{n,P} \Gamma_{n,PU}}{P_{n,S} \Gamma_{n,SU} + \frac{P_U}{SI} + B\omega_n N_0} \right) \\
 & = \log_2 \left(\sum_{j \in \{P,S\}} P_{n,j} \Gamma_{n,jU} + \frac{P_U}{SI} + B\omega_n N_0 \right) \\
 & \quad - \log_2 \left(P_{n,S} \Gamma_{n,SU} + \frac{P_U}{SI} + B\omega_n N_0 \right). \quad (35)
 \end{aligned}$$

For (35), $\log_2(\sum_{j \in \{P,S\}} P_{n,j} \Gamma_{n,jU} + [P_U/SI] + B\omega_n N_0)$ is non-convex with respect to variables $P_{n,P}$, $P_{n,S}$, P_U , and ω_n . Therefore, its convex upper bound is obtained by using the FTE in the r th iteration

$$\begin{aligned}
 & \log_2 \left(\sum_{j \in \{P,S\}} P_{n,j} \Gamma_{n,jU} + \frac{P_U}{SI} + B\omega_n N_0 \right) \\
 & \leq \log_2 \left(\sum_{j \in \{P,S\}} P_{n,j}^r \Gamma_{n,jU} + \frac{P_U^r}{SI} + B\omega_n^r N_0 \right) \\
 & \quad + \frac{\sum_{j \in \{P,S\}} \Gamma_{n,jU} (P_{n,j} - P_{n,j}^r) + \frac{P_U - P_U^r}{SI} + N_0 B (\omega_n - \omega_n^r)}{\ln 2 \left(\sum_{j \in \{P,S\}} P_{n,j}^r \Gamma_{n,jU} + \frac{P_U^r}{SI} + B\omega_n^r N_0 \right)} \\
 & \triangleq f_{R_{n,P1}}^r. \quad (36)
 \end{aligned}$$

The LHS expression of constraint $C_{1''}$ is nonconcave with respect to P_U and ω_n , we rewrite it into the following form:

$$\omega_n B \log_2 \left(P_{n,P} \Gamma_{n,P,U} + \frac{P_U}{SI} + B \omega_n N_0 \right) - \omega_n B \log_2 \left(\frac{P_U}{SI} + B \omega_n N_0 \right) \geq R_{n,th}. \quad (37)$$

For (37), we expand the concave lower bound of the second expression on the LHS of the inequality as

$$\log_2 \left(\frac{P_U}{SI} + B \omega_n N_0 \right) \leq \log_2 \left(\frac{P_U^r}{SI} + B \omega_n^r N_0 \right) + \frac{(P_U - P_U^r)/SI + (\omega_n - \omega_n^r) N_0 B}{\ln 2 \left(\frac{P_U^r}{SI} + B \omega_n^r N_0 \right)} \triangleq f_{R_{n,P2}}^r. \quad (38)$$

Therefore, problem (P4) can be transformed into the following convex optimization problem:

$$\begin{aligned} (\mathbf{P5}) : & \max_{\omega_n, P_{n,j}, P_U, \eta_{n,U}, \eta_{n,B}, \xi} \xi \\ \text{s.t. } & C_2, C_3, C_5, C_6 \\ & C_{01'} : f_{\xi}^r \leq \ln(\omega_n B) + \ln(\eta_{n,U}) \\ & C_{01''} : \eta_{n,U} \leq f_{\omega_n,U}^r \\ & C_{02'} : f_{\xi}^r \leq \ln(\omega_n B) + \ln(\eta_{n,B}) \\ & C_{02''} : \eta_{n,B} \leq f_{\omega_n,B}^r \\ & C_8 : \ln(\omega_n B) + \ln(\varphi) \leq \ln(R_{n,th}) \\ & C_9 : \varphi \geq f_{R_{n,P1}}^r \\ & -\log_2 \left(P_{n,S} \Gamma_{n,S,U} + \frac{P_U}{SI} + B \omega_n N_0 \right) \\ & C_{10} : \omega_n B \log_2 \left(P_{n,P} \Gamma_{n,P,U} + \frac{P_U}{SI} + B \omega_n N_0 \right) \\ & -f_{R_{n,P2}}^r \geq R_{n,th}. \end{aligned} \quad (39)$$

B. Altitude Optimization Subproblem

By fixing the power allocation variables $P_{n,j}$, P_U , and the bandwidth allocation variable ω_n , the problem (P3) can be transformed into

$$(\mathbf{P6}) : \max_{h_U, \xi} \xi \quad \text{s.t. } C_{01}, C_{02}, C_{1'}, C_{1''}, C_7. \quad (40)$$

(P6) is nonconvex since expressions on the RHS of constraints C_{01} , C_{02} , $C_{1'}$ are nonconcave with respect to the variable h_U , and the expression on the LHS of the constraint $C_{1'}$ is nonconvex with respect to h_U .

For constraint C_{01} , we introduce an auxiliary variable $\lambda_{n,jU}$ that satisfies the following condition:

$$\lambda_{n,jU} \leq \Gamma_{n,jU} \quad \forall n \in \{1, 2, \dots, N\}, j \in \{P, S\}. \quad (41)$$

The constraint C_{01} can be expressed as the following two constraints:

$$C_{01'} : \xi \leq \omega_n B \log_2 \left(1 + \frac{\sum_{j \in \{P,S\}} P_{n,j} \lambda_{n,jU}}{\frac{P_U}{SI} + B \omega_n N_0} \right) \quad (42)$$

$$\frac{1}{\lambda_{n,jU}} \geq \frac{D_{n,jU}^2}{\sigma_{NLoS} + \sigma_{LoS} a_0 \left(\frac{180}{\pi} \tan^{-1} \left(\frac{h_U}{d_{n,jU}} \right) - 15 \right)^{b_0}}. \quad (43)$$

According to [17, Lemma 1], we know that the RHS of the expression (43) is convex with respect to h_U , but it violates the DCP rule. We introduce another auxiliary variable $\mu_{n,jU}$ that satisfies

$$15 \leq \mu_{n,jU} \leq \frac{180}{\pi} \tan^{-1} \left(\frac{h_U}{d_{n,jU}} \right). \quad (44)$$

Then, (43) can be rewritten as

$$\frac{1}{\lambda_{n,jU}} \geq \frac{D_{n,jU}^2}{\sigma_{NLoS} + \sigma_{LoS} a_0 (\mu_{n,jU} - 15)^{b_0}}. \quad (45)$$

Although the RHS expression of (44) is concave with respect to h_U , it does not obey the DCP rule. By introducing the auxiliary variable κ satisfying

$$C_{7'} : \kappa \geq \frac{1}{h_U}. \quad (46)$$

Constraint (44) can be rewritten as

$$15 \leq \mu_{n,jU} \leq \frac{180}{\pi} \tan^{-1} \left(\frac{1}{d_{n,jU} \kappa} \right). \quad (47)$$

Since $\tan^{-1}(1/d_{n,jU} \kappa)$ is convex with respect to κ , we can get its concave lower bound by using FTE as

$$\tan^{-1} \left(\frac{1}{d_{n,jU} \kappa} \right) \geq \tan^{-1} \left(\frac{1}{d_{n,jU} \kappa^r} \right) - \frac{d_{n,jU} (\kappa - \kappa^r)}{1 + d_{n,jU}^2 (\kappa^r)^2} \triangleq f_{n,jU}^r. \quad (48)$$

Similarly, $(1/\lambda_{n,jU})$ is convex with respect to $\lambda_{n,jU}$, thus its concave lower bound can be given by using FTE as follows:

$$\frac{1}{\lambda_{n,jU}} \geq \frac{1}{\lambda_{n,jU}^r} - \frac{\lambda_{n,jU} - \lambda_{n,jU}^r}{(\lambda_{n,jU}^r)^2} \triangleq f_{\lambda,jU}^r. \quad (49)$$

Thus, we obtain constraints $C_{01'}$ and $C_{01''}$, respectively, as follows:

$$C_{01'} : \frac{d_{n,jU}^2 + h_U^2}{\sigma_{NLoS} + \sigma_{LoS} a_0 (\mu_{n,jU} - 15)^{b_0}} \leq f_{\lambda,jU}^r \quad (50)$$

and

$$C_{01''} : 15 \leq \mu_{n,jU} \leq \frac{180}{\pi} f_{n,jU}^r. \quad (51)$$

By introducing auxiliary variables $\lambda_{n,UB}$, $\mu_{n,UB}$, constraint C_{02} can be transformed in the same way as constraint C_{01}

$$C_{02''} : \xi \leq \omega_n B \log_2 \left(1 + \frac{P_U \lambda_{n,UB}}{B \omega_n N_0} \right) \quad (52)$$

and

$$C_{02'} : \frac{d_{n,UB}^2 + h_U^2}{\sigma_{NLoS} + \sigma_{LoS} a_0 (\mu_{n,UB} - 15)^{b_0}} \leq f_{\lambda,UB}^r \quad (53)$$

and

$$C_{02''} : 15 \leq \mu_{n,UB} \leq \frac{180}{\pi} f_{n,UB}^r. \quad (54)$$

For constraint $C_{1''}$, we introduce an auxiliary variable that satisfies $t_{n,jU} \geq \Gamma_{n,jU} \quad \forall n \in \{1, 2, \dots, N\}, j \in \{P, S\}$ and rewrite it as follows:

$$C_{11} : \omega_n B \log_2 \left(\sum_{j \in \{P, S\}} P_{n,j} t_{n,jU} + \frac{P_U}{SI} + B \omega_n N_0 \right) - \omega_n B \log_2 \left(P_{n,S} \lambda_{n,SU} + \frac{P_U}{SI} + B \omega_n N_0 \right) \leq R_{n,th} \quad (55)$$

and

$$\begin{aligned} & \ln(t_{n,jU}) + \ln\left(d_{n,jU}^2 + \frac{1}{\kappa^2}\right) \\ & \geq \ln\left(\sigma_{\text{NLoS}} + \sigma_{\text{LoS}} a_0 \left(\frac{180}{\pi} \tan^{-1}\left(\frac{h_U}{d_{n,jU}}\right) - 15\right)^{b_0}\right). \end{aligned} \quad (56)$$

Since the expression $\ln(d_{n,jU}^2 + [1/\kappa^2])$ is convex with respect to the variable κ , and the expression on the RHS of the inequality is concave with respect to h_U , we can obtain their concave lower bound and convex upper bound by using FTE, respectively. Then (56) can be reformulated as follows:

$$\begin{aligned} C_{12} : & \ln(t_{n,jU}) + \ln\left(d_{n,jU}^2 + \frac{1}{(\kappa^r)^2}\right) - \frac{2(\kappa - \kappa^r)}{d_{n,jU}^2(\kappa^r)^3 + \kappa^r} \\ & \geq \ln\left(\sigma_{\text{NLoS}} + \sigma_{\text{LoS}} a_0 \left(\frac{180}{\pi} \tan^{-1}\left(\frac{h_U^r}{d_{n,jU}}\right) - 15\right)^{b_0}\right) \\ & \quad + \frac{\frac{180}{\pi} \sigma_{\text{LoS}} a_0 \left(\frac{180}{\pi} \tan^{-1}\left(\frac{h_U^r}{d_{n,jU}}\right) - 15\right)^{b_0-1} \frac{d_{n,jU}(h_U - h_U^r)}{d_{n,jU}^2 + (h_U^r)^2}}{\sigma_{\text{NLoS}} + \sigma_{\text{LoS}} a_0 \left(\frac{180}{\pi} \tan^{-1}\left(\frac{h_U^r}{d_{n,jU}}\right) - 15\right)^{b_0}}. \end{aligned} \quad (57)$$

Therefore, we can rewrite problem (P6) at the r th iteration as a convex optimization problem as follows:

$$\begin{aligned} (\text{P7}) : & \max_{h_U, \{\mu_{n,jU}\}, \{\mu_{n,UB}\}, \{\lambda_{n,jU}\}, \{\lambda_{n,UB}\}, \kappa, \xi} \xi \\ \text{s.t.} & C_{01'}, C_{01''}, C_{01'''}, C_{02'}, C_{02''}, C_{02'''}, C_{11}, C_{12}, C_7, C_7' \\ & C_{21} : \omega_n B \log_2 \left(1 + \frac{P_{n,P} \lambda_{n,PU}}{\frac{P_U}{SI} + B \omega_n N_0} \right) \geq R_{n,th}. \end{aligned} \quad (58)$$

C. Power Splitting Scheme

With the ability of uplink RSMA to achieve the entire capacity region boundary, CR-RSMA transmission aims to maximize the channel capacity of high-energy users (SUs) while ensuring the QoS performance of low-energy users (PUs). After optimizing the bandwidth allocation, transmit power, and UAV's altitude in the first stage, we design an RE-prioritized scheduling scheme to perform the rate splitting operation for each pair, and the related power allocation factor α_n is designed as follows:

Since the PU can satisfy the QoS constraint if it occupies the entire channel bandwidth $\omega_n B$ but cannot satisfy the QoS constraint if it decodes the signals of all SUs as interference,

i.e., constraints $C_{01'}$ and $C_{01''}$. These above expressions can be further simplified as

$$\tau_n < \gamma_{n,PU} \leq \tau_n(1 + \gamma_{n,SU}) \quad \forall n \quad (59)$$

where $\tau_n = 2^{(R_{n,th}/\omega_n B)} - 1$.

In this case, the SU needs to perform a rate-splitting operation, and in order to ensure that the PU meets the QoS requirements and maximizes SU's achievable rate, the power splitting factor α_n should satisfy the following conditions:

$$\omega_n B \log_2 \left(1 + \frac{\gamma_{n,PU}}{(1 - \alpha_n) \gamma_{n,SU} + 1} \right) = R_{n,th}. \quad (60)$$

Thus, the closed-form expression of the power splitting factor can be obtained by

$$\alpha_n = 1 - \frac{\gamma_{n,PU} - \tau_n}{\gamma_{n,SU} \tau_n}. \quad (61)$$

Substituting (61) into (7) obtains the achievable rates for the n th pair of SU as

$$\begin{aligned} R_{n,SU} = & \omega_n B \log_2 \left(\frac{\tau_n(1 + \gamma_{n,PU} + \gamma_{n,SU})}{\gamma_{n,PU}(1 + \tau_n)} \right) \\ & + \omega_n B \log_2 \left(\frac{\gamma_{n,PU}}{\tau_n} \right). \end{aligned} \quad (62)$$

D. Overall Algorithm

We propose an iteration-based EURT algorithm according to the analysis of the above three sections. First, we alternately optimize the bandwidth allocation factor, UAV's altitude, and transmit power to get a suboptimal solution of problem (P3). Then, we obtain the optimal power splitting factor by the RE-prioritized CR-RSMA scheme. Define $\pi(h_U, \omega_n)$, $\hat{\pi}_{P5}(h_U, \omega_n)$, and $\hat{\pi}_{P7}(h_U, \omega_n)$ as the value of the objective function of problem (P3), (P5), and (P7), respectively. The convergence of the algorithm can be proved as follows:

$$\begin{aligned} \pi(h_U^r, \omega_n^r) & \stackrel{(a)}{=} \hat{\pi}_{P5}(h_U^r, \omega_n^r) \\ & \stackrel{(b)}{\leq} \hat{\pi}_{P5}(h_U^{r+1}, \omega_n^r) \stackrel{(c)}{\leq} \pi(h_U^{r+1}, \omega_n^r) \\ & \stackrel{(d)}{=} \hat{\pi}_{P7}(h_U^{r+1}, \omega_n^r) \stackrel{(e)}{\leq} \hat{\pi}_{P7}(h_U^{r+1}, \omega_n^{r+1}) \\ & \stackrel{(f)}{\leq} \pi(h_U^{r+1}, \omega_n^{r+1}) \end{aligned} \quad (63)$$

where (a) and (d) are valid as the approximation provided by FET is tightly coupled to the original function in the vicinity of the point of expansion; (b) and (e) are available since the optimal solutions of altitude and bandwidth allocation optimization subproblems are found in problems (P5) and (P7), respectively; Since solutions for problems (P5) and (P7), respectively, result in optimal concave lower bounds on the original issue (P3), (c) and (f) are valid.

Algorithm 1 summarizes the solution process of the overall algorithm. To iteratively optimize both subproblem (P5) and (P7) by using the SCA method, we need to initialize the optimization variables. The initial UAV altitude is set to the minimum flight altitude, i.e., $h_U^0 = h_{\min} =$

Algorithm 1 Overall EURT Algorithm

```

1: Initialization:  $\omega_n^0, h_U^0$ ;  $r, a, b = 0, \zeta^0 = 0$ .
2: repeat
3:    $r = r + 1$ .
4:    $\omega_n^a = \omega_n^{a-1}, \xi^a = \xi^{a-1}$ .
5:   repeat
6:      $a = a + 1$ .
7:     Fix  $h_U^{a-1}$  to solve problem (P5), obtain the variable
        $\omega_n^a$  and the objective value  $\xi_B^a$ .
8:   until  $\frac{\xi_B^a - \xi_B^{a-1}}{\xi_B^a} < \epsilon$ .
9:   repeat
10:     $b = b + 1$ ;  $h_U^{b-1} = h_U^a, \xi_A^{b-1} = \xi_B^a$ .
11:    Fix  $\omega_n^{b-1}$  to solve problem (P7), obtain the variable
        $h_U^b$  and the objective value  $\xi_A^b$ .
12:  until  $\frac{\xi_A^b - \xi_A^{b-1}}{\xi_A^b} < \epsilon$ .
13:  Let  $\xi^r = \xi_A^b$ .
14: until  $\frac{\xi^r - \xi^{r-1}}{\xi^r} < \epsilon$ .
15: Update the optimization variables  $\omega_n^r, h_U^r$ .
16: for  $n = 1:N$  do
17:   Calculate  $\gamma_{n,1U}, \gamma_{n,2U}$ , and  $\tau_n$ .
18:   if  $\gamma_{n,1U} \geq \tau_n(1 + \gamma_{n,2U})$  then
19:      $\alpha_n = 0$ .
20:   else if  $\tau_n < \gamma_{n,1U} \leq \tau_n(1 + \gamma_{n,2U})$  then
21:      $\alpha_n = 1 - \frac{\gamma_{n,1U} - \tau_n}{\gamma_{n,2U} - \tau_n}$ .
22:   else
23:      $\alpha_n = 1$ .
24:   end if
25: end for

```

$\min\{d_{n,jU}, d_{UB}\} \tan(\pi/12) \quad \forall n$. For the optimization problem (P5), ω_n^0 is set as $\omega_1 = \omega_2 = \dots = \omega_N = 1/N$, and we set

$$\xi^0 = \omega_n^0 B \log_2 \left(1 + \frac{\sum_{j=1}^2 P_{n,j} \Gamma_{n,jU}}{\omega_n^0 B N_0} \right). \quad (64)$$

For the optimization problem (P7), we set $\kappa^0 = 1/h_U^0$

$$\lambda_{n,jU}^0 = \frac{\sigma_{NLoS} + \sigma_{LoS} a_0 \left(\frac{180}{\pi} \tan^{-1} \left(\frac{h_U^0}{d_{n,jU}} \right) - 15 \right)^{b_0}}{d_{n,jU}^2 + (h_U^0)^2} \quad (65)$$

and

$$\lambda_{n,UB}^0 = \frac{\sigma_{NLoS} + \sigma_{LoS} a_0 \left(\frac{180}{\pi} \tan^{-1} \left(\frac{h_U^0}{d_{UB}} \right) - 15 \right)^{b_0}}{d_{UB}^2 + (h_U^0)^2}. \quad (66)$$

The complexity of Algorithm 1 is $O[N^3 \log(1/\epsilon)]$ since problems (P5) and (P7) are solved by the interior-point method. ϵ and N represent a determined convergence accuracy [39] and the overall amount of optimization variables in problems (P5) and (P7), respectively.

IV. SIMULATION RESULTS

In this section, numerical simulations are provided to demonstrate the effectiveness of the proposed EURT algorithm. We consider an emergency communication network

TABLE II
PARAMETER SETUP

Parameter	Value
Unit channel attenuation: β_0	-80 dB
Delay threshold: T_{th}	60 s
PU's QoS threshold: $R_{n,th}$	30 Mbps
Self-interference cancellation coefficient: SI	130 dB
Noise power spectrum density: N_0	-169 dBm/Hz
Total system bandwidth : B	5 MHz
Convergence accuracy: ϵ	1×10^{-3}
UAV total energy budget: E_U	120 KJ
Blade profile power: P_0	79.8563 w
Induced power: P_i	88.6279 w

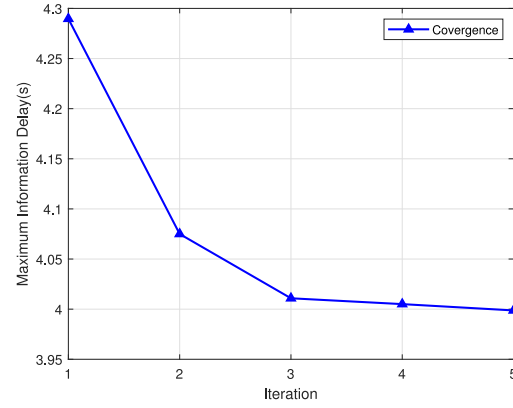


Fig. 4. Convergence of the EURT algorithm.

with limited users' RE and UAV's energy budget. We use the urban environment in [17] for our simulation analysis, where $a_0 = 0.6$, $b_0 = 0.11$, $\eta_{LoS} = 1$, and $\eta_{NLoS} = 20$. We assume that $M = 4$ AUs are trapped in the affected area, and their coordinates are (250, 0, 0), (100, 10, 0), (-200, 62, 0), and (-84, 22, 0), respectively. The available BS is located at coordinate (1000, 0, 0). The initial position of the drone is at (0, 0, 100). The UAV's hovering position is located at $x_U = 0$ m and $y_U = 0$ m. The length of the packet to be transmitted for each user is 200 Mbit. The remaining parameters are summarized in Table II.

Fig. 4 shows that as the number of iterations increases, the maximum transmission delay monotonically decreases and gradually remains stable. Therefore, the proposed algorithm is guaranteed to converge to the given error ϵ within a finite number of iterations.

The user delay fairness factor τ defined through Jain's fairness index [40] is expressed as

$$\tau = \frac{\left(\sum_{i=1}^M t_i \right)^2}{M \sum_{i=1}^M t_i^2} \quad (67)$$

where t_i represents the transmission delay of AU_i . Jain's fairness factor is an index that can quantitatively reflect the fairness of the system. It can be seen that the fairness index is always continuous and between 0 and 1, and it can be applied to variables of any number and measurement scale. If all t_i values are the same, the fairness index τ is 1, which means the system has maximum fairness, i.e., 100% fair. In contrast, if only one value of is greater than 0 and the rest are 0, the

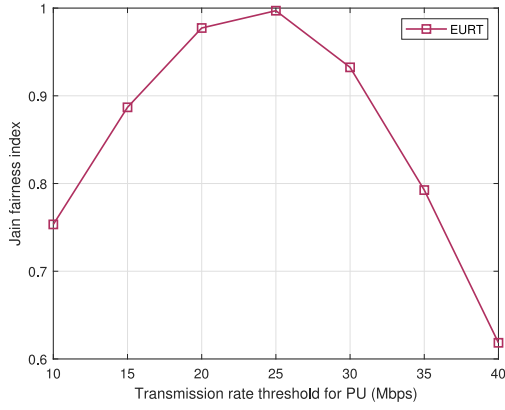


Fig. 5. Tradeoff between optimal performance and network fairness.

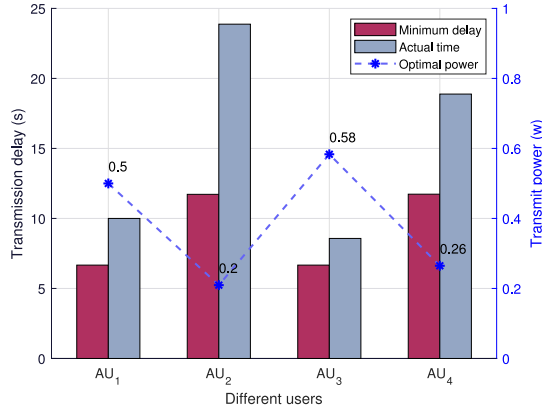


Fig. 6. Comparison between the optimal transmission delay and the actual usage time of different user's device when the RE is 15 KJ.

fairness index is $(1/M)$, which means that the system is unfair to $1 - (1/M)$ of the users.

Fig. 5 shows the balance between the optimal transmission performance of PU and the fairness of the network. It can be observed that with the increase of $R_{n,th}$, the fairness of the network will first monotonically increase. When the value of $R_{n,th}$ is 25 Mb/s, the network can achieve optimal fairness under the condition of satisfying PU performance, and then the fairness will monotonically decrease. This is because the proposed algorithm will improve SU's performance as much as possible when the QoS requirement of PU is low, while sacrificing some SU performance to meet the performance requirements of PU when the QoS threshold of PU is high, which is consistent with our previous analysis.

Fig. 6 compares the optimal transmission delay achieved by the proposed EURT algorithm with the actual usage time when the RE is 15 KJ, and shows the optimal transmit power of different users' devices. It can be seen that the optimal transmission delay of each user is less than the actual available time of the user's device, which shows that the proposed EURT algorithm can complete data transmission before the RE of users' device is exhausted.

To demonstrate the effectiveness of the proposed EURT algorithm, we design the following two benchmark algorithms:

- 1) *Resource Allocation Algorithm for CR Inspired NOMA (CR-NOMA)*: In this algorithm, the users are paired

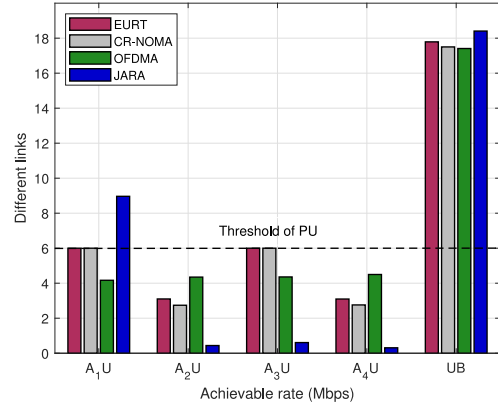


Fig. 7. Achievable rate versus different links.

by the same approach as the EURT algorithm. The UAV is operated with FD mode, the available system spectrum is divided into orthogonal subcarriers and each pair of users occupy one subcarrier to transmission. The bandwidth allocation, transmit power, and UAV altitude are jointly optimized to minimize the maximum delay among all users while guaranteeing the QoS constraints.

- 2) *Resource Allocation Algorithm for Orthogonal Frequency Division Multiple Access (OFDMA)*: In this algorithm, the available system spectrum is divided into M orthogonal subcarriers with the same bandwidth, and each user occupies one subcarrier to send signal to the UAV relay simultaneously. The optimal user's power allocation, and drone hovering altitude are designed to achieve max-min transmission delay fairness.
- 3) *Joint Altitude and Resource Allocation (JARA) Algorithm [17]*: In this algorithm, all users transmit data to the UAV relay within the same frequency band using uplink NOMA. The total available spectrum of the system is divided into two orthogonal sub-bandwidths, and the first-hop and second-hop of the relay link occupy one frequency band, respectively. The maximum throughput of the backhaul link is achieved by designing the optimal bandwidth allocation factor, users' transmit power, and UAV's altitude.

Fig. 7 shows achievable rates for different user access and UAV backhaul links with different algorithms. We set the QoS threshold for PU as 6 Mb/s. The RE values for all AUs are specified as follows: $E_1 = 15$ KJ, $E_2 = 25$ KJ, $E_3 = 20$ KJ, and $E_4 = 30$ KJ. It can be seen that the proposed EURT strategy maximizes the channel capacity of SU while guaranteeing the QoS threshold of PU, since it is able to adaptively adjust the interference to PU by splitting the rate of SU. Furthermore, CR-NOMA is capable of achieving the same achievable rate for PU, but the achievable rate for SU is lower than that in EURT. This is due to the fact that NOMA can only achieve two corner points of the MAC capacity region, whereas RSMA is able to realize all the boundary points, as illustrated in Fig. 3. However, the OFDMA algorithm does not meet low-RE users' performance threshold. It is noteworthy that the JARA algorithm can significantly improve the achievable rate of the A_1U with the lowest RE. The main reason is that

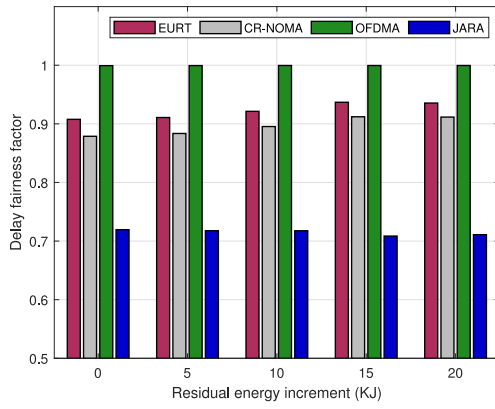


Fig. 8. Delay fairness versus RE increment for different algorithms.

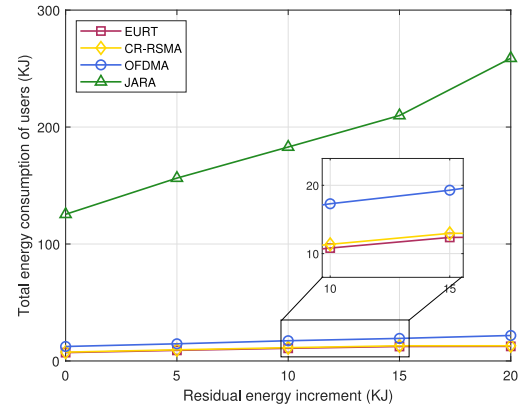


Fig. 10. EC versus RE increment for different algorithms.

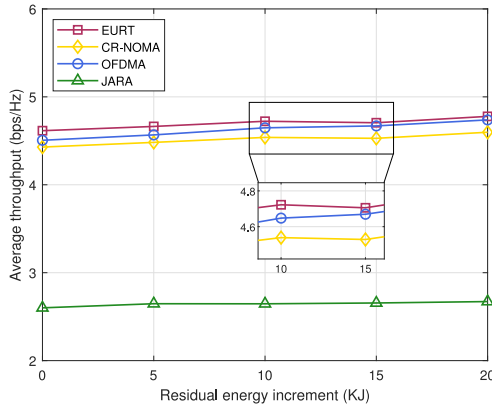


Fig. 9. Throughput versus RE increment for different algorithms.

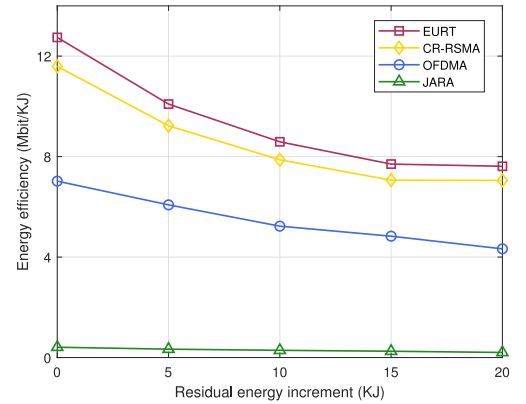


Fig. 11. EE versus RE increment for different algorithms.

NOMA transmission causes earlier decoded users to regard the signals of the remaining undecoded users as interference, while the last decoded users will not suffer from interchannel interference. When RE is limited, it is difficult for users to achieve network fairness by adjusting transmit power.

Fig. 8 compares the user delay fairness of the four algorithms with different RE, in which the JARA algorithm has the worst fairness as the energy gap between different users increases. Moreover, compared to the CR-NOMA and JARA algorithms, the proposed EURT algorithm is capable of achieving better fairness as it can adaptively adjust the threshold of the PU to facilitate a tradeoff between performance and fairness for each pair of users. Although OFDMA-based algorithms can achieve optimal fairness by optimizing the bandwidth allocation of subcarriers, UAV's altitude, and transmit power, low-RE users may not satisfy the QoS requirements of PUs, as we discussed in Fig. 7.

It can be seen from the average throughput in Fig. 9, the proposed EURT algorithm always outperforms the other three baseline algorithms in terms of the average throughput of the network as the user's RE increases. Note that the JARA algorithm has the worst network throughput performance, due to its lack of an appropriate user pairing scheme. This results in the user who undergoes SIC decoding first being subjected to interference from all subsequent users. When RE is limited, it becomes challenging for the early decoding users to achieve fairness through adjustment of the transmission power.

Figs. 10 and 11 compare the total user EC and EE of four algorithms under different increments of RE, respectively. It is observed that with the increase in RE increments, the EC of all four algorithms monotonically increases, while the EE monotonically decreases. Compared to the other three algorithms, the proposed EURT algorithm prioritizes the performance of users with low RE and reduces interference from high-RE users. Consequently, the proposed EURT is able to achieve lower EC and higher EE, as well as fairness similar to that of the orthogonal multiple access algorithms.

Furthermore, to prove the validity of our proposed RE-based pairing strategy, Figs. 12–14 compare performance metrics with regard to delay, EC, and EE of SUs under different pairing strategies: 1) the EURT-CSI strategy is a CSI-based pairing method that pairs the worst CSI user with the best CSI user, pairs the second-worst user with the second-best user, and sets the user with the worst CSI as a PU; 2) the EURT-P1 strategy denotes that the smallest RE user is paired with the largest RE user, where the user with the smaller RE is the PU; and 3) the EURT-P2 strategy pairs users in ascending order of RE value and considers the user with the lower RE as the PU.

Fig. 12 shows that the proposed energy-aware approach enables SUs to achieve lower transmission latency compared to the other three pairing approaches. In particular, the gain of the proposed method is more apparent when the RE increment is low.

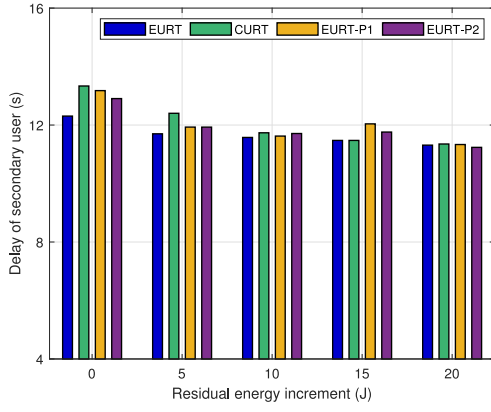


Fig. 12. SU's delay versus RE increment for different pairing methods.

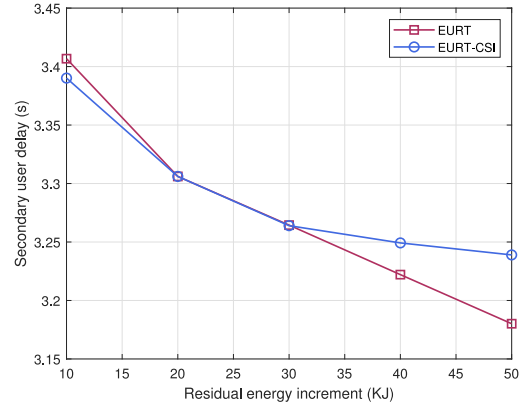


Fig. 15. Delay versus RE increment for different scheduling policies.

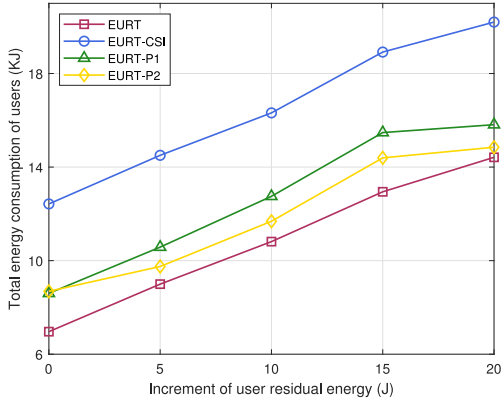


Fig. 13. EC versus RE increment for different pairing methods.

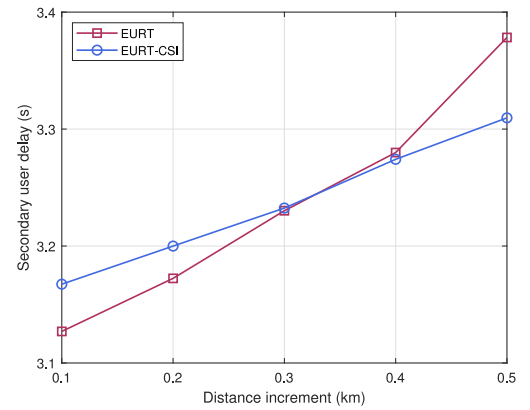


Fig. 16. Delay versus distance increment for different scheduling policies.

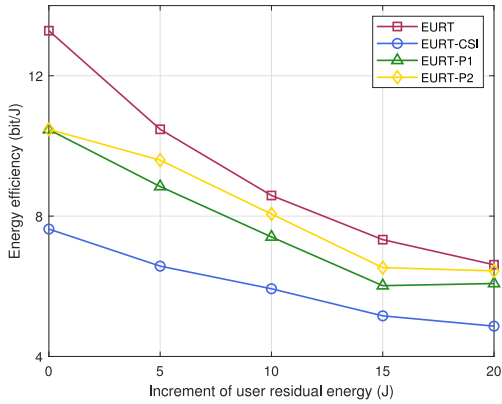


Fig. 14. EE versus RE increment for different pairing methods.

Figs. 13 and 14 show that our proposed EURT scheme realizes lower EC and higher EE for different RE increments than other pairing strategies, respectively. In contrast, the CSI-based pairing strategy has the worst EC and EE performance. It can be inferred that since the energy-based pairing strategy always treats the M users with the lowest RE as the PUs and prioritizes their performance, the high-RE users need to adjust their transmit power to reduce their interference to the PUs, which reduces the system EC. Therefore, our proposed EURT scheme has better network performance regarding delay, EC, and EE when users' energies are limited.

Moreover, to evaluate the impact of RE-aware and CSI-aware scheduling strategies on fairness, we designed an evaluation framework to quantify the tradeoff between the two strategies, and the evaluation results are shown in Figs. 15 and 16. For simplicity of analysis, we consider the case where the number of users is $M = 2$. First, we fix the positions of user-1 and user-2 and set their coordinates as $(-0.08, -0.02)$ and $(0.3, 0)$, and set the RE of both user-1 and user-2 as 10 KJ. Then, we compare the SU delays of the two strategies by increasing the increment of the RE of user-2, which is regarded as a measure of fairness. Fig. 15 shows that when the RE increment is less than 20 KJ, our proposed RE-aware strategy achieves better fairness than CSI-aware strategy. In contrast, the CSI-aware strategy can achieve better fairness when the RE increment exceeds 30 KJ. Similarly, we set the RE of user-1 and user-2 as 15 and 25 KJ and their coordinates as $(-0.08, -0.02)$ and $(0.1, 0)$. We compare the SU delay achieved by the two strategies by increasing the distance increment of user-2. Fig. 16 shows that when the distance increment is less than 0.3km, the CSI-aware strategy can achieve better fairness than the RE-aware strategy. In comparison, the RE-aware strategy has better fairness when the distance increment is more than 0.3km.

Therefore, we infer that the RE-aware scheduling policy achieves better fairness when the user's RE gap is high. In comparison, the CSI-aware scheduling policy has better

fairness when users are far away from each other. Meanwhile, the above conclusion can also be used as a measurement basis for the system to choose different scheduling strategies.

V. CONCLUSION

In this article, we investigated a max-min delay fairness problem for UAV-assisted emergency relay networks with energy-constrained users. To tradeoff the optimal transmission delay and network fairness, we proposed an EURT algorithm for subcarrier band allocation, transmit power allocation, and UAV altitude design. To mitigate the network delay unfairness caused by the limited RE of users, we designed an RE-prioritized CR-RSMA scheduling policy that pairs all users two-by-two according to RE, and minimizes the SU's delay while satisfying the PUs' QoS threshold of each pair. Compared to the other algorithms and pairing methods in the literature, the simulation results demonstrated that the proposed EURT algorithm can achieve better performance in terms of throughput, delay, EC, and EE, as well as an ideal tradeoff between delay and fairness.

In the proposed network, the mobility of users under realistic disaster scenarios has not been fully explored. In future work, we will integrate offline optimization and online learning to provide reliable dynamic time-varying channel prediction for UAVs.

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harvesting, Age of Information, and distributionally robust optimization.

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